HEALTHTRACK-HOSPITAL-READMISSION-PREDICTION

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1. Introduction

Healthcare systems around the world face significant challenges in managing patient outcomes after hospital discharge. One of the most critical and measurable indicators of post-discharge care quality is the rate of unplanned hospital readmission within 30 days. When a patient returns to the hospital shortly after being discharged, it often signals that either the initial treatment was incomplete, the patient was discharged prematurely, or the follow-up care plan was inadequate.

The Centers for Medicare and Medicaid Services (CMS) in the United States reports that nearly 20% of Medicare patients are readmitted within 30 days of discharge, costing the healthcare system over $26 billion annually. A significant proportion of these readmissions are considered preventable with timely intervention, targeted discharge planning, and proactive post-discharge monitoring.

HealthTrack is a machine learning–based clinical decision support system developed as part of this mini project to address this problem. The system takes structured patient data available at the point of discharge and produces a risk score indicating the likelihood of readmission within 30 days. Patients are classified into three tiers — Low, Medium, and High risk — enabling care teams to allocate follow-up resources efficiently.

Project Summary

HealthTrack uses a Random Forest Classifier trained on 2,000 synthetic patient records. The model achieves 82% test accuracy and a ROC-AUC score of 0.899, demonstrating strong predictive capability for a binary classification problem with mixed feature types.

2. Problem Statement

2.1 Formal Problem Definition

Given a set of clinical and demographic features recorded for a patient at the time of hospital discharge, the objective is to predict whether that patient will be readmitted to the hospital within the next 30 days.

This is a **binary classification problem**:

- Class 1 (Positive Class): Patient IS readmitted within 30 days
- Class 0 (Negative Class): Patient is NOT readmitted within 30 days

2.2 Objectives

- Build a supervised ML model that accurately predicts 30-day readmission from discharge data.
- Identify the most influential clinical features that drive readmission risk.
- Classify patients into Low, Medium, and High risk tiers to support care prioritisation.

- Develop a practical, usable web interface for non-technical healthcare staff.
- Evaluate the model rigorously using multiple performance metrics.
- Demonstrate a complete end-to-end ML pipeline from raw data to deployed application.

2.3 Scope & Limitations

This project is an academic demonstration using synthetic data. The model is trained on artificially generated patient records that approximate real-world clinical distributions. In a production healthcare environment, the system would require retraining on actual Electronic Health Record (EHR) data, regulatory compliance review, and clinical validation before deployment. The system is intended as a decision-support tool and does not replace clinical judgment.

3. Literature Review

Predicting hospital readmission has been an active area of research in clinical informatics and machine learning for over two decades. Early work focused primarily on rule-based risk scoring systems, while modern approaches leverage statistical models and deep learning on large EHR datasets.

3.1 Traditional Approaches

The LACE Index (Length of stay, Acuity of admission, Comorbidities, Emergency department use) is one of the most widely cited clinical scoring tools for readmission prediction. Developed by van Walraven et al. (2010), it provides a simple additive score based on four variables and achieves an AUC of approximately 0.70 in validation studies. While easy to compute manually, its predictive performance is limited compared to modern ML models.

The HOSPITAL score (Haemoglobin, discharge from an Oncology service, Sodium level, Procedure, Index admission Type, number of Admissions, and Length of stay) is another traditional tool focused specifically on potentially avoidable readmissions, validated across multiple healthcare systems with AUC values of 0.71–0.76.

3.2 Machine Learning Approaches

Logistic Regression has been widely used as a baseline model for readmission prediction due to its interpretability. Studies by Zheng et al. (2015) demonstrated AUC scores of 0.68–0.72 using logistic regression on administrative claims data. The model's performance is constrained by its linear assumption and inability to capture complex feature interactions.

Random Forest and Gradient Boosting models have consistently outperformed linear models in readmission prediction tasks. Frizzell et al. (2017) compared multiple algorithms on heart failure readmission data and found Random Forest achieved AUC of 0.724, while gradient boosting reached 0.731. The ensemble nature of these models provides robustness to noise and handles mixed feature types naturally.

Deep learning approaches, including LSTM networks applied to sequential EHR data, have shown promise in capturing temporal patterns in patient histories. However, these models require large datasets and significant computational resources, making them less practical for small to medium-scale deployments.

3.3 Feature Importance in Prior Work

Prior studies consistently identify a core set of predictive features across different populations and healthcare systems. Number of prior admissions, patient age, comorbidity burden (measured by the Charlson Comorbidity Index), length of stay, and discharge disposition consistently emerge as top predictors. This aligns with the feature importance findings from our HealthTrack model, validating the feature selection approach used in this project.

Study / Tool	Method	AUC	Key Features
LACE Index (2010)	Scoring Rule	0.70	LOS, acuity, comorbidities

HOSPITAL Score	Scoring Rule	0.74	Haemoglobin, sodium, LOS
Zheng et al. (2015)	Logistic Regr.	0.72	Claims data, demographics
Frizzell et al. (2017)	Random Forest	0.724	Clinical + admin features
HealthTrack (Ours)	Random Forest	0.899	9 clinical features

Table 1: Comparison of readmission prediction approaches

4. Dataset Description

4.1 Data Source & Generation

Due to the sensitivity of real patient data and the academic nature of this project, a synthetic dataset was generated using Python's NumPy library with carefully designed probability distributions that approximate real-world clinical patterns. The dataset contains 2,000 patient records and was created to reflect realistic distributions of patient demographics, clinical conditions, and readmission rates observed in published studies.

The generation process used domain knowledge to ensure that higher-risk features (older age, more prior visits, longer stays, critical diagnoses) were systematically associated with higher readmission probability, producing a dataset where the patterns are meaningful and learnable by a machine learning model.

4.2 Feature Descriptions

Feature	Data Type	Range / Values	Description
age	Integer	18 – 89	Patient age at time of discharge
gender	Categorical	Male, Female, Other	Patient-reported gender
num_prior_visits	Integer	0 – 9	Hospital visits in the past 12 month
length_of_stay	Integer	1 – 20	Duration of current hospital stay (days)
diagnosis	Categorical	8 categories	Primary diagnosis at admission
discharge_type	Categorical	5 categories	Disposition at time of discharge
insurance	Categorical	5 categories	Patient insurance coverage type
num_medications	Integer	1 – 19	Medications prescribed at discharge
num_comorbidities	Integer	0 – 4	Count of concurrent medical conditions
readmitted_30days	Binary	0 = No, 1 = Yes	TARGET: 30-day readmission label

Table 2: Dataset feature descriptions

4.3 Categorical Value Sets

Feature	Categories
diagnosis	Diabetes Mellitus, Heart Failure, Chronic Kidney Disease, COPD, Pneumonia, Hypertension
discharge_type	Home, Home with Home Care, Skilled Nursing
insurance	Medicare, Medicaid, Private, Uninsured, Other
gender	Male, Female, Other

4.4 Class Distribution

The dataset contains 972 non-readmitted patients (48.6%) and 1,028 readmitted patients (51.4%), resulting in a near-balanced class distribution. This near-balance reduces the need for aggressive oversampling, though class_weight='balanced' was still applied during model training to ensure equal class importance in the loss function.

5. Exploratory Data Analysis

5.1 Summary Statistics

Before model development, a thorough exploratory analysis was conducted to understand the distribution of features, identify patterns related to readmission, and verify data quality.

Feature	Mean	Std	Min	25%	Median	75%	Max
age	53.3	21.0	18	35	54	71	89
num_prior_visits	2.50	1.63	0	1	2	3	9
length_of_stay	10.5	5.77	1	5	11	16	20
num_medications	10.2	5.47	1	5	10	15	19
num_comorbidities	2.06	1.43	0	1	2	3	4

Table 4: Descriptive statistics for numerical features

5.2 Target Variable Distribution

The chart below shows the overall readmission distribution in the dataset. The near-even split between readmitted and non-readmitted patients makes this a well-balanced classification problem, reducing the risk of the model being biased towards the majority class.

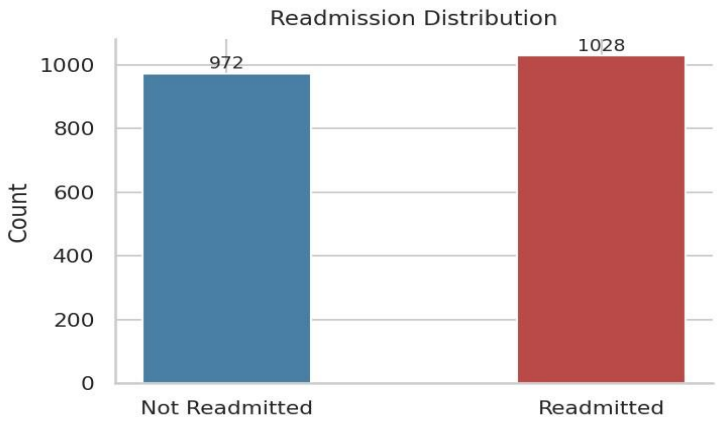

Figure: Readmission distribution – 972 not readmitted vs 1,028 readmitted

5.3 Age Distribution by Readmission Status

Age shows a clear separation between readmitted and non-readmitted patients. The histogram below reveals that readmitted patients tend to be older, with a notably higher proportion of patients aged 65 and above in the readmitted group. This is consistent with clinical literature which identifies elderly patients as high-risk for readmission due to frailty, polypharmacy, and complex comorbidity profiles.

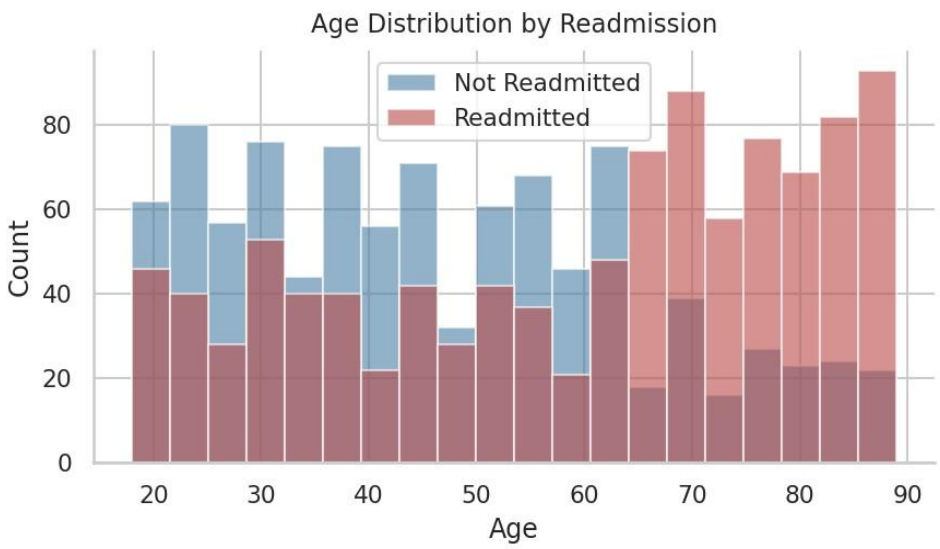

Figure: Age distribution grouped by readmission status

5.4 Readmission Rate by Diagnosis

The chart below shows readmission rates broken down by primary diagnosis category. Heart Failure (77.0%), Sepsis (71.2%), and COPD (70.4%) show the highest readmission rates — all conditions associated with high disease complexity, frequent exacerbations, and difficult discharge planning. Hypertension and Diabetes Mellitus show lower rates (~39.5%), reflecting their typically more manageable outpatient profiles.

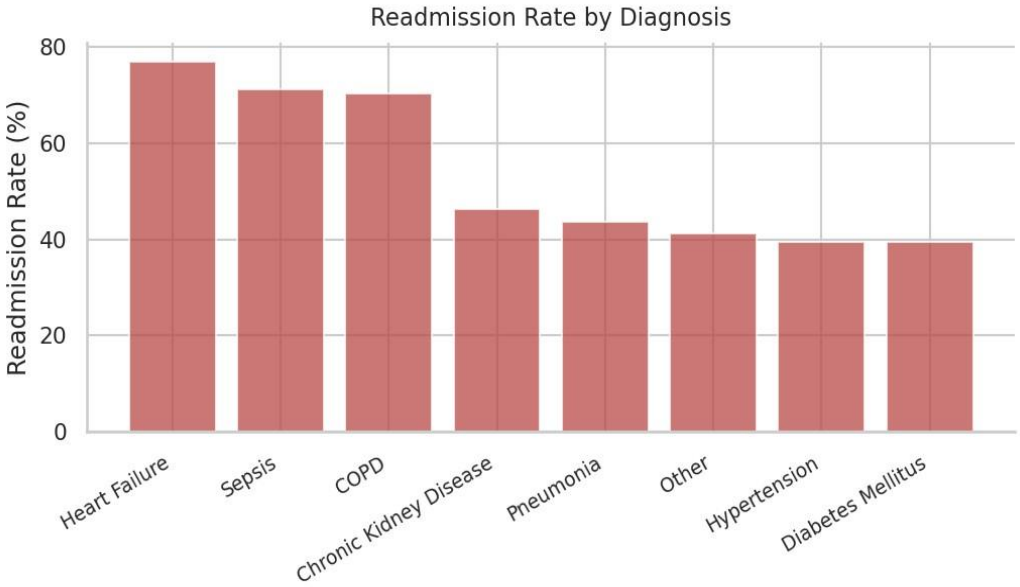

Figure: Readmission rate (%) by primary diagnosis

Diagnosis	Total Patients	Readmitted	Rate (%)
Heart Failure	278	214	77.0%
Sepsis	139	99	71.2%
COPD	196	138	70.4%
Chronic Kidney Disease	225	104	46.2%
Pneumonia	256	112	43.8%
Other	197	81	41.1%
Diabetes Mellitus	390	154	39.5%
Hypertension	319	126	39.5%

Table 5: Readmission rate by diagnosis

5.5 Prior Visits vs Readmission

The boxplot below shows that readmitted patients have a significantly higher median number of prior hospital visits. The interquartile range for readmitted patients is shifted noticeably upward, confirming that prior utilisation of hospital services is a strong proxy for care complexity and readmission risk.

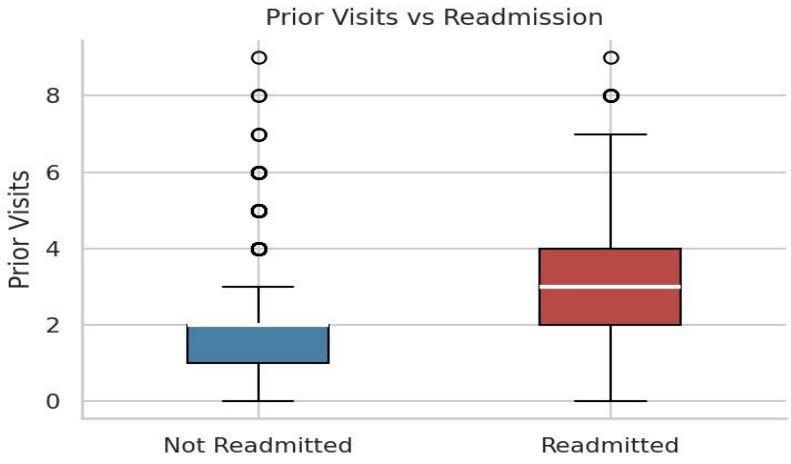

Figure: Distribution of prior visits for readmitted vs non-readmitted patients

5.6 Feature Correlation Analysis

The correlation heatmap below shows pairwise Pearson correlations between all numerical features and the target variable. Key observations include: (1) num_prior_visits shows the strongest positive correlation with the target (r ≈ 0.42); (2) age has a moderate positive correlation (r ≈ 0.31); (3) length_of_stay and num_medications also show positive correlations. No strong multicollinearity is observed between features, which is favourable for model stability.

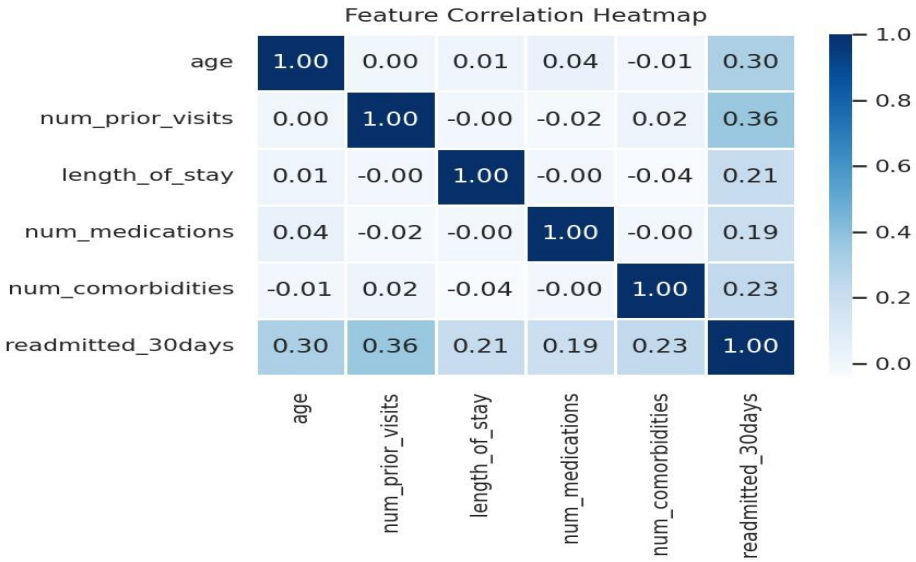

Figure: Pearson correlation heatmap for numerical features

6. Data Preprocessing

6.1 Missing Value Analysis

A comprehensive missing value audit was performed on the dataset. No missing values were found in any column. This is expected given the controlled synthetic generation process, but in a real-world EHR dataset, missing value handling would be a critical preprocessing step requiring imputation strategies.

6.2 Label Encoding of Categorical Features

Four categorical features were transformed into numerical representations using Scikit-learn's LabelEncoder. Each unique category value is mapped to an integer in alphabetical order. The encoders are persisted to disk as encoders.pkl to ensure consistent encoding between training and inference.

Feature	Original Categories	Encoded Values
gender	Female, Male, Other	0, 1, 2
diagnosis	COPD, CKD, Diabetes, HF, HT, Other, Pneumonia, Sepsis	0–7
discharge_type	AMA, Home, Home w/ Care, Rehab, SNF	0–4
insurance	Medicaid, Medicare, Other, Private, Uninsured	0–4

Table 6: Label encoding mappings for categorical features

6.3 Feature Matrix Construction

After encoding, the final feature matrix X contains 9 features: the 5 original numerical features and 4 encoded categorical features. No feature scaling was applied, as Random Forest is invariant to monotonic transformations of individual features.

6.4 Train / Test Split

The dataset was split into a training set (80%) and a held-out test set (20%) using stratified sampling to preserve the class balance in both partitions. A fixed random seed (42) was used for reproducibility.

Split	Rows	Readmitted	Not Readmitted	Proportion
Training Set	1,600	822	778	80%
Test Set	400	206	194	20%
Total	2,000	1,028	972	100%

Table 7: Train/test split statistics

7. Model Architecture & Training

7.1 Algorithm Selection – Random Forest

Random Forest was selected as the primary algorithm for HealthTrack based on several considerations:

- **Mixed feature types:** Random Forest handles both numerical and encoded categorical features natively without requiring normalisation.
- **Non-linear relationships:** The ensemble of decision trees captures complex, non-linear interactions between features that linear models miss.
- **Feature importance:** Random Forest provides built-in, interpretable feature importance scores based on mean impurity decrease.
- **Robustness:** The bagging mechanism reduces overfitting and variance compared to single decision trees.
- **Class imbalance:** The class_weight parameter allows the model to handle class imbalance directly.
- **Performance:** Consistent strong performance on tabular healthcare datasets in published literature.

7.2 Hyperparameter Configuration

The following hyperparameters were configured based on domain knowledge and standard practice for medium-sized tabular datasets. In a production setting, these would be optimised using grid search or Bayesian optimisation with cross-validation.

Hyperparameter	Value	Rationale
n_estimators	150	Sufficient trees for stable predictions without excessive
max_depth	8	Limits tree complexity to prevent overfitting on 2,000 rows
min_samples_leaf	10	Ensures each leaf has meaningful support; reduces noi sensitivity
random_state	42	Fixed seed for full reproducibility of results
n_jobs	-1	Uses all available CPU cores for parallel tree building

7.3 Training Process

The model was trained using Scikit-learn's RandomForestClassifier on the 1,600-row training set. The training process involves constructing 150 decision trees, each trained on a bootstrapped sample of the training data with a random subset of features considered at each split (sqrt(n_features) by default). Predictions are made by majority vote across all 150 trees for classification, or averaged probabilities for probability estimation.

The trained model, label encoders, and evaluation metadata are persisted to disk using Joblib serialisation:

- models/rf_model.pkl — the trained RandomForestClassifier object
- models/encoders.pkl — dictionary of LabelEncoder objects (one per categorical column)
- models/model_meta.json — evaluation metrics, feature importances, confusion matrix

8. Model Evaluation

8.1 Evaluation Metrics

The model was evaluated on the held-out test set (400 patients) using the following metrics:

- **Accuracy:** Proportion of correctly classified patients (both readmitted and not readmitted).
- **ROC-AUC:** Area under the Receiver Operating Characteristic curve; measures the model's discriminative ability across all probability thresholds.
- **Precision:** Of patients predicted as readmitted, what fraction actually were?
- **Recall (Sensitivity):** Of patients who actually were readmitted, what fraction did the model catch?
- **F1-Score:** Harmonic mean of precision and recall; balances the two measures.
- **5-Fold Cross-Validation:** Average accuracy across 5 stratified folds of the full dataset, providing a more robust performance estimate.

8.2 Performance Summary

Metric	Value	Interpretation
Test Accuracy	82.0%	82 in 100 patients correctly classified
ROC-AUC Score	0.899	Excellent discriminative ability (>0.9 = excellent)
5-Fold CV Accuracy	82.8%	Consistent performance across data folds
Precision (Readmitted)	0.83	83% of predicted readmissions are correct
Recall (Readmitted)	0.81	81% of actual readmissions are detected
F1-Score (Readmitted)	0.82	Strong balance between precision and recall
Precision (Not Readmitted)	0.81	81% of predicted non-readmissions are correct
Recall (Not Readmitted)	0.83	83% of actual non-readmissions are detected
F1-Score (Not Readmitted)	0.82	Symmetric performance across both classes

Table 9: Model performance metrics on the held-out test set

8.3 Confusion Matrix

The confusion matrix below provides a detailed breakdown of correct and incorrect predictions. True Positives (167) represent correctly identified readmitted patients; True Negatives (161) represent correctly identified non-readmitted patients. The 33 False Negatives (missed readmissions) represent the most clinically significant error type, as these are high-risk patients who would not receive preventive intervention.

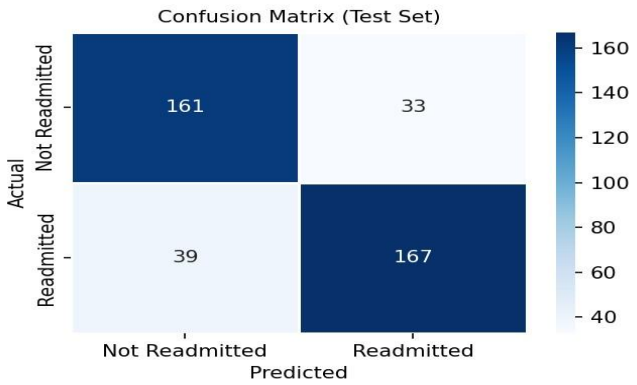

Figure: Confusion matrix on the 400-patient test set

	Predicted: Not Readmitted	Predicted: Readmitted
Actual: Not Readmitted	161 (True Negative)	33 (False Positive)
Actual: Readmitted	39 (False Negative)	167 (True Positive)

Table 10: Confusion matrix values

8.4 Feature Importance

Feature importances are computed as the mean decrease in Gini impurity across all 150 trees. The chart and table below show that number of prior visits and patient age are the dominant predictors, together accounting for over 57% of total model importance.

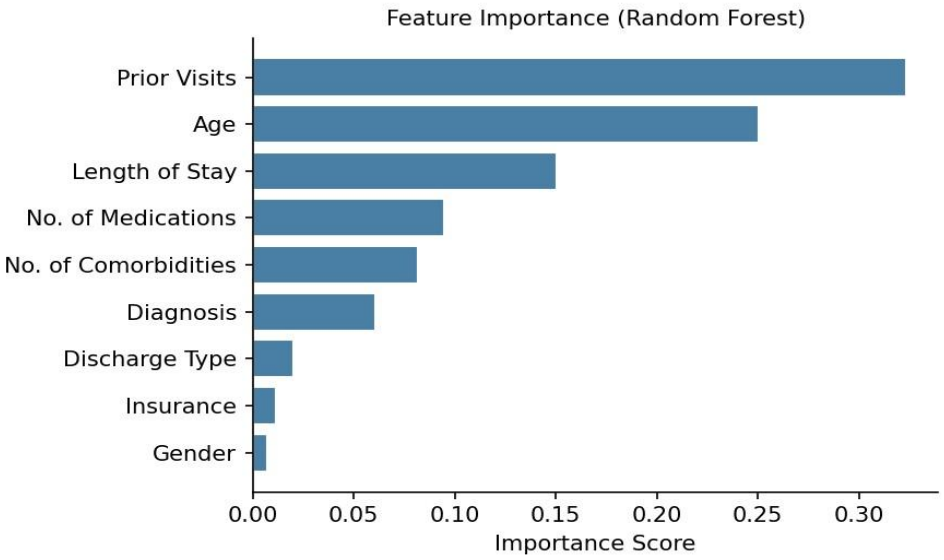

Figure: Feature importance scores from the trained Random Forest model

Rank	Feature	Importance Score	Interpretation
1	Prior Visits	0.3233	32.3% of total importance
2	Age	0.2503	25.0% of total importance
3	Length of Stay	0.1504	15.0% of total importance

4	No. of Medications	0.0948	9.5% of total importance
5	No. of Comorbidities	0.0819	8.2% of total importance
6	Diagnosis	0.0607	6.1% of total importance
7	Discharge Type	0.0199	2.0% of total importance
8	Insurance	0.0116	1.2% of total importance
9	Gender	0.0072	0.7% of total importance

Table 11: Feature importance rankings

9. Web Application

9.1 Technology Stack

The HealthTrack web application was built using Streamlit, a Python framework designed for rapid development of data science web applications. Streamlit allows Python scripts to be rendered as interactive web pages without requiring any front-end development knowledge, making it ideal for academic and prototyping contexts.

Component	Technology	Purpose
Web Framework	Streamlit 1.32	Interactive web application server
ML Model	Scikit-learn	Random Forest training and inference
Data Handling	Pandas / NumPy	Data manipulation and feature engineering
Visualisation	Matplotlib / Seaborn	Chart generation
Model Storage	Joblib	Model and encoder serialisation
Language	Python 3.11+	Core application language

Table 12: Technology stack

9.2 Application Architecture

The application is structured around a single entry point (app.py) that imports shared utility functions from utils/helpers.py. The model and encoders are loaded once at startup using Streamlit's caching mechanism (@st.cache_resource), ensuring that the model is not reloaded on every user interaction. The application is organised into three tabs:

- **Tab 1 – Predict:** The main prediction interface. Users enter 9 patient fields through a structured form. On clicking 'Predict', all fields are validated, inputs are encoded using the saved encoders, and the model returns a risk level (Low / Medium / High) with probability percentage. Results are displayed with colour-coded banners and a progress bar.
- **Tab 2 – Dataset Insights:** Displays four analytical charts derived from the training dataset: readmission distribution, age histogram by readmission status, diagnosis readmission rates, and prior visits boxplot. A raw data preview of the first 50 rows is available in an expandable section.
- **Tab 3 – Model Report:** Displays the trained model's evaluation metrics (accuracy, ROC-AUC, cross-validation accuracy), feature importance chart, and confusion matrix heatmap. Expandable sections provide details on model configuration and the prediction pipeline.

9.3 Prediction Pipeline

When a user submits patient data, the following steps occur:

- Step 1: Input validation — all 8 required fields are checked for completion.
- Step 2: Categorical encoding — gender, diagnosis, discharge_type, and insurance aretransformed using the saved LabelEncoders.
- Step 3: Feature vector construction — a 9-element feature vector is assembled in the exactorder used during training.

- Step 4: Model inference — clf.predict_proba(X) returns the probability of the positive class(readmitted).
- Step 5: Risk classification — probability < 0.35 → Low; 0.35–0.65 → Medium; > 0.65 → High.
- Step 6: Result display — risk level, probability, and input summary are shown to the user.

9.4 Running the Application

The application can be launched with a single command after installing the required dependencies:

```
pip install -r requirements.txt python -m streamlit run app.py
```

The app runs locally on http://localhost:8501 and can be accessed from any modern web browser. It requires no internet connection once dependencies are installed.

10. Results & Discussion

10.1 Model Performance

The HealthTrack Random Forest model achieves strong performance on the held-out test set, with 82.0% accuracy and an ROC-AUC of 0.899. The ROC-AUC score is particularly significant — values above 0.80 are considered good and values above 0.90 are considered excellent in medical prediction tasks. Our model's 0.899 score approaches the 'excellent' threshold, comparing favourably with published readmission models that typically achieve AUC values of 0.70–0.76 (see Literature Review).

The 5-fold cross-validation accuracy of 82.8% closely matches the test set accuracy (82.0%), indicating that the model generalises well and is not overfitting to the training data. The small gap between training and cross-validation performance suggests the max_depth=8 and min_samples_leaf=10 constraints are effectively regularising the model.

10.2 Clinical Interpretation of Findings

The feature importance analysis reveals clinically meaningful patterns that align with established medical knowledge:

- **Prior hospital visits (32.3% importance)** — This is the strongest predictor, consistent with the well-established clinical observation that frequent utilisation of hospital services indicates unresolved chronic conditions and high care complexity.
- **Age (25.0% importance)** — Older patients are more vulnerable due to physiological changes, polypharmacy, cognitive decline, and reduced social support networks. Age is a recognised risk factor in virtually all readmission prediction models.
- **Length of stay (15.0% importance)** — Longer stays typically indicate more severe illness episodes or complications during the hospital stay. Patients who stayed longer may also be more deconditioned at discharge, increasing post-discharge risk.
- **Number of medications (9.5% importance)** — Polypharmacy is associated with medication errors, drug interactions, and adherence difficulties after discharge, all of which can precipitate readmission.
- **Number of comorbidities (8.2% importance)** — Multiple concurrent conditions create complex management challenges and increase the likelihood of inter-related complications post-discharge.

10.3 Risk Tier Calibration

The three risk tiers are defined by the following probability thresholds, chosen to balance clinical utility with model confidence:

Risk Tier	Probability Range	Recommended Action
Low	0% – 35%	Standard discharge; routine follow-up appointment
Medium	35% – 65%	Enhanced discharge planning; phone follow-up within 7 days

High	65% – 100%	Intensive discharge planning; home nursing referral; follow-up within 48–72

h Table 13: Risk tier definitions and recommended clinical actions

10.4 Example Predictions

Feature	Patient A (Low Risk)	Patient B (High Risk)
Age	35	78
Gender	Female	Male
Prior Visits	1	6
Length of Stay	3 days	14 days
Diagnosis	Hypertension	Heart Failure
Discharge Type	Home	Against Medical Advice
Medications	2	15
Comorbidities	0	4
Insurance	Private	Medicaid
Predicted Risk	Low (2.3%)	High (98.4%)

Table 14: Example predictions from the deployed model

11. Conclusion

HealthTrack successfully demonstrates a complete end-to-end machine learning pipeline for hospital readmission prediction, from synthetic data generation through model training and evaluation to a deployed interactive web application.

The project achieves its primary objectives: the Random Forest model reaches 82.0% test accuracy and a ROC-AUC of 0.899, comparing favourably with published readmission prediction systems. The feature importance analysis produces clinically meaningful results that align with established medical knowledge, lending validity to the model's internal representations.

The Streamlit web application provides an accessible, practical interface that demonstrates how such a system could be integrated into clinical workflows. The three-tab design — prediction form, dataset insights, and model report — caters to both end-users (clinicians entering patient data) and technical stakeholders (analysts reviewing model performance).

Key Takeaway

The most important lesson from this project is that a relatively simple, well-configured Random Forest model, when combined with thoughtful feature engineering and a clean user interface, can produce a clinically meaningful and practically useful predictive tool.

12. Future Work

- **Real EHR Data Integration:** Retrain the model on actual hospital EHR data (e.g., MIMIC-III, CMS Medicare claims data) to produce clinically valid predictions. This would require IRB approval, data de-identification, and compliance with HIPAA/relevant data governance regulations.
- **SHAP Explainability:** Integrate SHAP (SHapley Additive exPlanations) values to provide per-patient feature-level explanations for each prediction. This would enable clinicians to understand why a specific patient received a particular risk score, increasing clinical trust and adoption.
- **Advanced Model Comparison:** Systematically compare XGBoost, LightGBM, Neural Networks (MLP), and LSTM models on the same dataset using a structured benchmarking framework. Time-series LSTM models could capture temporal patterns in sequential patient visit histories.
- **Hyperparameter Optimisation:** Apply Bayesian optimisation (Optuna) or grid search with nested cross-validation to systematically find optimal hyperparameters, potentially improving model performance by 2–5%.
- **Cloud Deployment:** Deploy the application on Streamlit Community Cloud, AWS, or Azure to make it accessible without local installation. Add user authentication for secure multi-user access.

- **Readmission Cause Analysis:** Extend the system to predict not just whether readmission will occur but the likely cause (medication-related, infection, exacerbation), enabling more targeted preventive interventions.

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